

Cover Letter

[AUTHOR INPUT REQUIRED: submission date]

Dear Editor-in-Chief,

We submit the Original Research manuscript entitled “Return-aware proximal policy optimization with pointer attention for multi-resource mountain-road unmanned aerial vehicle inspection planning” for consideration in Engineering Applications of Artificial Intelligence.

The engineering problem is online planning for fixed, priority-weighted mountain-road inspection points under simultaneous energy, distance, mission-time, dynamics, and depot-return constraints. The artificial-intelligence contribution is a priority-aware PPO–Pointer actor–critic with a deterministic return-aware legal-action mask. The engineering contribution is a locked, reproducible evaluation spanning unseen procedural mountain maps, independent public Copernicus DSM regions, classical optimisers, robustness layers, and component ablations.

The primary evidence is deliberately bounded. PPO+Pointer substantially exceeds an attention-free fixed-slot PPO, but it does not exceed attention-matched A2C+Pointer on confirmatory safe weighted coverage. ACO, SA, and MILP achieve higher coverage in several comparisons at much higher online computation. The strongest component result is the large loss caused by removing the return reserve. We make no claim of real-flight validation, safety certification, out-of-range scale generalisation, or universal PPO superiority.

The manuscript is formatted as a single-column, double-anonymised Original Research submission. The abstract is below 250 words, five keywords and five Highlights are supplied, and the package includes data/code availability, funding, competing-interest, authorship, and generative-AI declarations or explicit author placeholders. Scientific figures were redrawn only from frozen Source Data; no generative image tool was used.

[AUTHOR INPUT REQUIRED: confirm that the work is original, is not under consideration elsewhere, and has been approved by all authors.]

[AUTHOR INPUT REQUIRED: identify any related manuscripts, preprints, or prior conference versions and explain overlap.]

Suggested reviewers: [AUTHOR INPUT REQUIRED: names, institutions, institutional emails, expertise, and conflict checks, if the journal requests suggestions.]

Thank you for considering this manuscript.

Sincerely,

[AUTHOR INPUT REQUIRED: corresponding author name and contact information]